PROGRESS PRESENTATION 2

AI-Based Smart Swimming Pool Monitoring and Maintenance System

Group R26-IT-143 · Four integrated AI modules delivered on one live dashboard

M H M Saajith IT22156310 — Crowd-Aware Maintenance

N M Shafan IT22248480 — Drowning Detection

Yogarasa Nilogini IT22247254 — Water Quality Prediction

M Y S Arsalan IT22277336 — Garbage Intrusion Detection

Supervisor: Prof. Anuradha Karunasena Co-Supervisor: Mr. Samadhi Rathnayake Academic Year 2026

0:00 - 0:45

Problem, Stakeholders and the Measurable Pain Point

Swimmers

- Silent drowning is difficult to notice
- No way to check water safety before entry
- Debris reduces hygiene and comfort

Lifeguards

- Attention degrades across long shifts
- Occlusion hides distress in busy water
- Manual occupancy counting is error-prone

Operators

- Cleaning follows a calendar, not real usage
- Chemistry is checked periodically only
- No single view of safety and hygiene

The measurable pain point

- Safety supervision, water chemistry and maintenance are run today as three disconnected manual processes.
- Every decision is reactive: staff act after a problem is already visible rather than before it develops.
- Maintenance effort is scheduled by calendar, so it is either wasted on a quiet day or applied too late on a busy one.
- No operator view links crowd load to water degradation or to safety risk, so the three signals are never reasoned about together.

0:45 - 1:30

Research Gap and Objectives

Gap in existing work

- Pool safety literature treats drowning detection, water monitoring and litter detection as separate systems
- Crowd analytics is mature in retail and transport but rarely applied to pool maintenance scheduling
- No reported system reasons across all four streams in one decision layer

Our position

- The contribution is the integration layer, not a single new detector
- Crowd load feeds maintenance urgency and raises alert priority
- Every claim is supported by measurement on held-out data

	Objective	Status at PP2
O1	Detect swimmers and derive bather load for dynamic maintenance	Delivered with pool-region filtering
O2	Classify water quality as Safe, Warning or Critical	Delivered with a trained classifier
O3	Detect drowning-like behaviour from CCTV in real time	Delivered with persistence-based alerting
O4	Detect garbage intrusion without alerting on pool objects	Delivered with class disambiguation
O5	Integrate all modules into one decision engine and dashboard	Delivered as a single shared pipeline
O6	Evaluate with standard metrics and document limitations	Delivered, 27 automated checks passing

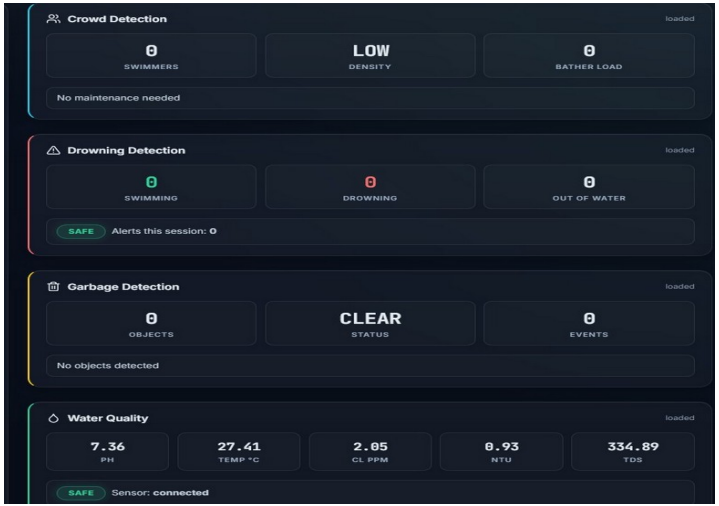
1:30 - 2:30

Solution Overview and System Architecture

Data Acquisition	Shared camera stream · water sensor array
AI Inference	Crowd · Drowning · Garbage · Water quality
Event Processing	Tracking, bather load, temporal alert persistence
Decision Intelligence	Cross-module fusion: risk level and maintenance urgency
Visualisation	Unified dashboard, alert feed, session report

Why the decision layer matters

- High bather load with a Warning water reading raises urgency earlier than either signal alone
- A drowning alert during high density is escalated, because staff confirmation will be slower



Live dashboard: four modules reporting simultaneously

INDIVIDUAL CONTRIBUTION · COMPONENT 1

Crowd-Aware Maintenance Scheduling

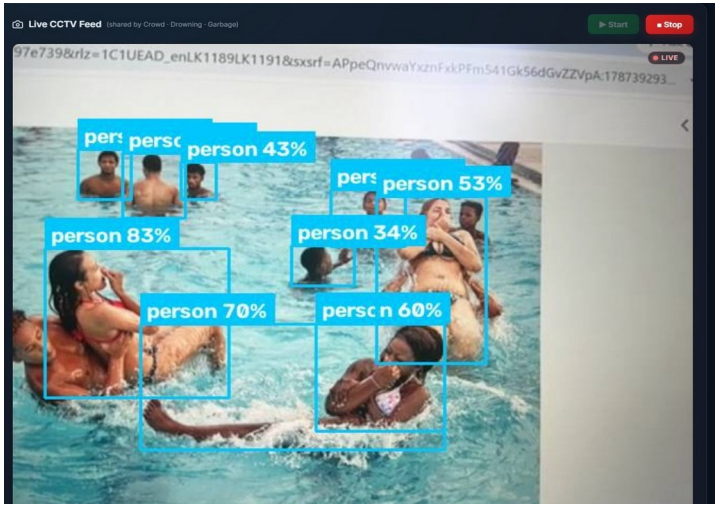
M H M Saajith IT22156310 Occupancy analytics and maintenance triggering

What this module does

- Detects people in the camera view and filters them by a configurable pool region, so only swimmers in the water are counted
- Accumulates bather load in person-hours from the counted occupancy over time
- Raises maintenance actions when accumulated load passes defined thresholds, and classifies density against pool capacity

Individual contribution at PP2

- Replaced a narrow single-class detector with a general person detector after measured comparison on identical footage
- Added pool-region filtering so people on the deck no longer inflate bather load
- Added rolling-median smoothing and overlap-threshold tuning to remove duplicate and flickering counts
- Capacity thresholds derived from published water-area-per-bather guidance rather than arbitrary numbers



Live detection output from the running system

0.794
mAP@50 (BASELINE MODEL)

9
DETECTED IN TEST SCENE

ROI
IN-WATER FILTERING

INDIVIDUAL CONTRIBUTION · COMPONENT 2

AI-Based Water Quality Prediction

Yogarasa Nilogini	IT22247254	Sensor pipeline and quality classification

What this module does

- Reads five water parameters continuously: pH, temperature, chlorine, turbidity and total dissolved solids
- Classifies the current water state as Safe, Warning or Critical using a trained gradient-boosting model
- Publishes status to the dashboard on an independent thread and raises an alert only when the state changes

Individual contribution at PP2

- Built the sensor acquisition path with tolerant parsing so malformed readings are skipped rather than crashing the loop
- Trained and deployed the classifier with its scaler and label encoder as a reproducible artefact set
- Added a hardware-free simulation mode so the pipeline and dashboard are demonstrable without the sensor rig
- Reported the dataset's perfect separability openly as a property of the data, not as a performance claim

5
SENSOR PARAMETERS

3
QUALITY CLASSES

2 s
UPDATE INTERVAL

INDIVIDUAL CONTRIBUTION · COMPONENT 3

Swimmer Behaviour and Drowning Detection

N M Shafan	IT22248480	Safety monitoring and emergency alerting

What this module does

- Classifies each detected swimmer as Swimming, Drowning, or Person out of water from the shared camera frame
- Requires the drowning state to persist across consecutive frames before an alert is raised
- Escalates the alert to the dashboard with an on-screen banner and an entry in the alert history

Individual contribution at PP2

- Trained the three-class behaviour model and produced the strongest evaluation scores in the project
- Designed the frame-persistence rule that suppresses single-frame false alarms without delaying genuine alerts
- Prioritised recall in threshold selection, on the basis that a missed event is more costly than a false alarm
- Documented that the training data is staged, so the model detects drowning-like posture rather than verified incidents

0.907
mAP@50

0.883
RECALL

3
BEHAVIOUR CLASSES

INDIVIDUAL CONTRIBUTION · COMPONENT 4

Garbage Intrusion Detection

M Y S Arsalan IT22277336 Hygiene monitoring and false-alarm control

What this module does

- Detects foreign objects on and around the water surface across six trained object classes
- Separates genuine litter from legitimate pool objects, so balls and leaves are detected but never raise a hygiene alert
- Reports intrusion events to the dashboard and contributes to the maintenance urgency calculation

Individual contribution at PP2

- Trained the six-class detector and defined the class-disambiguation rule that suppresses false hygiene alerts
- Packaged the detector as an importable module so it runs inside the shared pipeline and standalone
- Identified under-training and a small test split as limitations, with longer GPU training recorded as future work
- Attributed the third-party dataset in line with its licence terms

0.903	0.853	6
mAP@50	PRECISION	OBJECT CLASSES

2:30 - 3:30

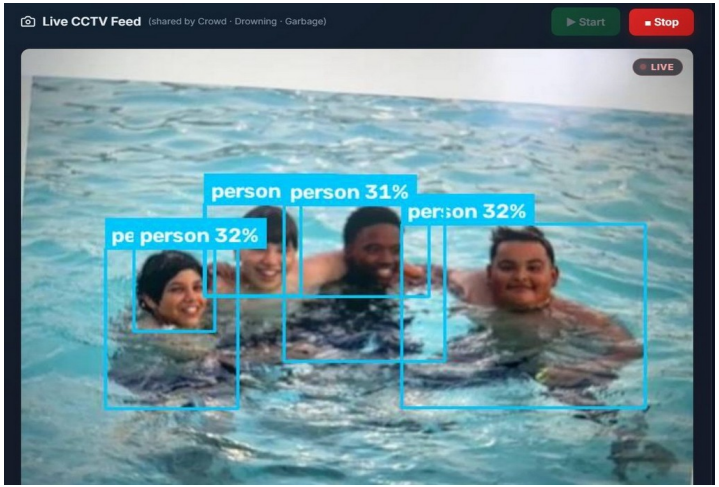
Integration and Completion Status

4 / 4	27 / 27	18.3	1
MODULES INTEGRATED	SYSTEM CHECKS PASSED	FRAMES PER SECOND, CPU	SHARED CAMERA STREAM

Component	Owner	Runtime status
Crowd-aware maintenance	M H M Saajith	Loaded and running
Water quality prediction	Yogarasa Nilogini	Loaded and running
Drowning detection	N M Shafan	Loaded and running
Garbage intrusion	M Y S Arsalan	Loaded and running

Engineering evidence

- One frame is processed by three vision modules in the same loop; water quality runs independently
- Missing model, camera or sensor is reported on the dashboard instead of stopping the system
- Each member's module runs standalone as well as inside the shared dashboard



Shared camera frame processed by the vision modules

3:30 - 4:00

Evaluation Evidence

Measured on held-out test data. The crowd test split contains 148 images and 1,482 annotated instances.

Module	Model	Precision	Recall	mAP@50	mAP@50-95
Drowning detection	Three-class behaviour model	0.877	0.883	0.907	0.594
Garbage intrusion	Six-class object model	0.853	0.878	0.903	0.393
Crowd - swimmer v1	Single-class, single source	0.777	0.760	0.794	0.338
Crowd - swimmer v2	Single-class, combined sources	0.785	0.765	0.793	0.334

Why these metrics

- Recall is prioritised for safety modules: a missed event costs more than a false alarm
- mAP@50-95 is reported beside mAP@50 so box-localisation quality is not hidden behind a friendlier number

Water quality classifier

- Five input features, three output classes, deployed with its scaler and encoder
- Separates the augmented dataset perfectly, which is reported as a property of the data rather than as an accuracy claim

RESEARCH STRENGTH

Three Findings That Changed Our Design

01

More data did not improve accuracy

A larger combined dataset scored no better than the smaller single-source set (mAP@50 0.793 against 0.794) on the same test split. Combining sources introduced differences in camera angle, lighting and annotation style, so added volume did not add generalisation.

02

A general detector beat our domain-specific model

On identical live footage the custom swimmer model reported one person while a general person detector reported nine. For counting, the breadth of large-scale pretraining outweighed narrow fine-tuning on a limited dataset.

03

Threshold selection is a measurable trade-off

Lowering the confidence threshold recovered missed swimmers but produced duplicate detections elsewhere. Tightening the overlap threshold removed the duplicates, and the pair was fixed as the operating point.

4:00 - 4:30

Non-Functional Requirements, Ethics and Security

Requirement	Demonstrated result
Real-time performance	18.3 frames per second on a laptop CPU with no GPU
Reliability	Bounded camera reconnection; missing model or sensor reported, not fatal
Usability	One operator view, responsive from desktop to mobile browser
Maintainability	Each module is a self-contained package with its own entry point
Portability	Camera and sensor sources selected by configuration, no code change

Ethical, security and social considerations

- Privacy by design: the system counts and classifies behaviour but does not identify individuals, and raw footage is not retained.
- Human oversight: outputs are decision support for lifeguards and operators, and do not replace supervision.
- Honest scope: drowning behaviour is learned from staged footage, so the model detects drowning-like posture rather than verified real incidents.
- Licensing: third-party datasets are used under their stated terms and are attributed in the report.

Limitations and Validity Threats

Limitation	Effect on validity	Mitigation applied
Drowning data is staged	Acted postures may differ from genuine distress	Scope stated explicitly; frame persistence reduces false alarms
Water dataset is synthetic	Perfect classification reflects separable data	Reported as a data property; retraining on logged data is future work
Occlusion in crowded water	Overlapping swimmers merged or missed	Overlap threshold tuned; rolling median smooths variation
Small test splits	Metric confidence intervals are wide	Test set sizes reported beside every metric
Single-site conditions	Results may not transfer to other pools	Input source configurable; multi-site validation is future work

A complete system without defensible evidence is still weak research. These limitations are reported because they shape how the results should be read.

4:30 - 5:00

Commercialisation and User Benefit

Target customers

- Hotels and resorts with guest pools
- Municipal and school facilities
- Gyms and residential complexes

Value delivered

- A continuous second observer for lifeguards
- Cleaning matched to actual bather load
- Earlier warning before water becomes unsafe

Revenue model

- Annual subscription per pool site
- Optional camera and sensor bundle
- Reporting tier for audit records

Phase	Activity	Outcome sought
Pilot	Deploy at a small number of local facilities at no licence cost	Operational data and reference sites
Partnership	Distribute through pool maintenance and construction providers	Reach without direct sales cost
Scale	Multi-site licensing for hotel groups and municipal operators	Recurring revenue across grouped sites

Cost advantage: the system runs on standard laptop hardware with commodity cameras, which is what makes smaller facilities addressable.

HANDOVER

Demonstration Roadmap

What the panel will see in the 30-minute demonstration, and what each step is evidence of.

	What we will show	What it demonstrates
1	Dashboard starts with all four modules reporting loaded	Integration is real, not four separate programs
2	Live feed with swimmers detected inside the pool region	Region-filtered counting, not naive person counting
3	A person outside the boundary excluded from the count	Bather load reflects people in the water only
4	Water values updating with a classified status	Sensor pipeline running independently of the camera
5	Drowning and garbage reporting on the same frame	One shared stream feeding three vision models
6	Decision engine raising urgency from combined signals	Cross-module reasoning, the core contribution
7	Generating the session report	Actionable operator output, not only a screen

Fallback prepared: if live camera conditions are poor, the same pipeline is demonstrated on recorded footage and in simulation mode.

Thank you

Questions, then the live demonstration

R26-IT-143 · Four integrated AI modules · One unified dashboard · 27 of 27 system checks passing